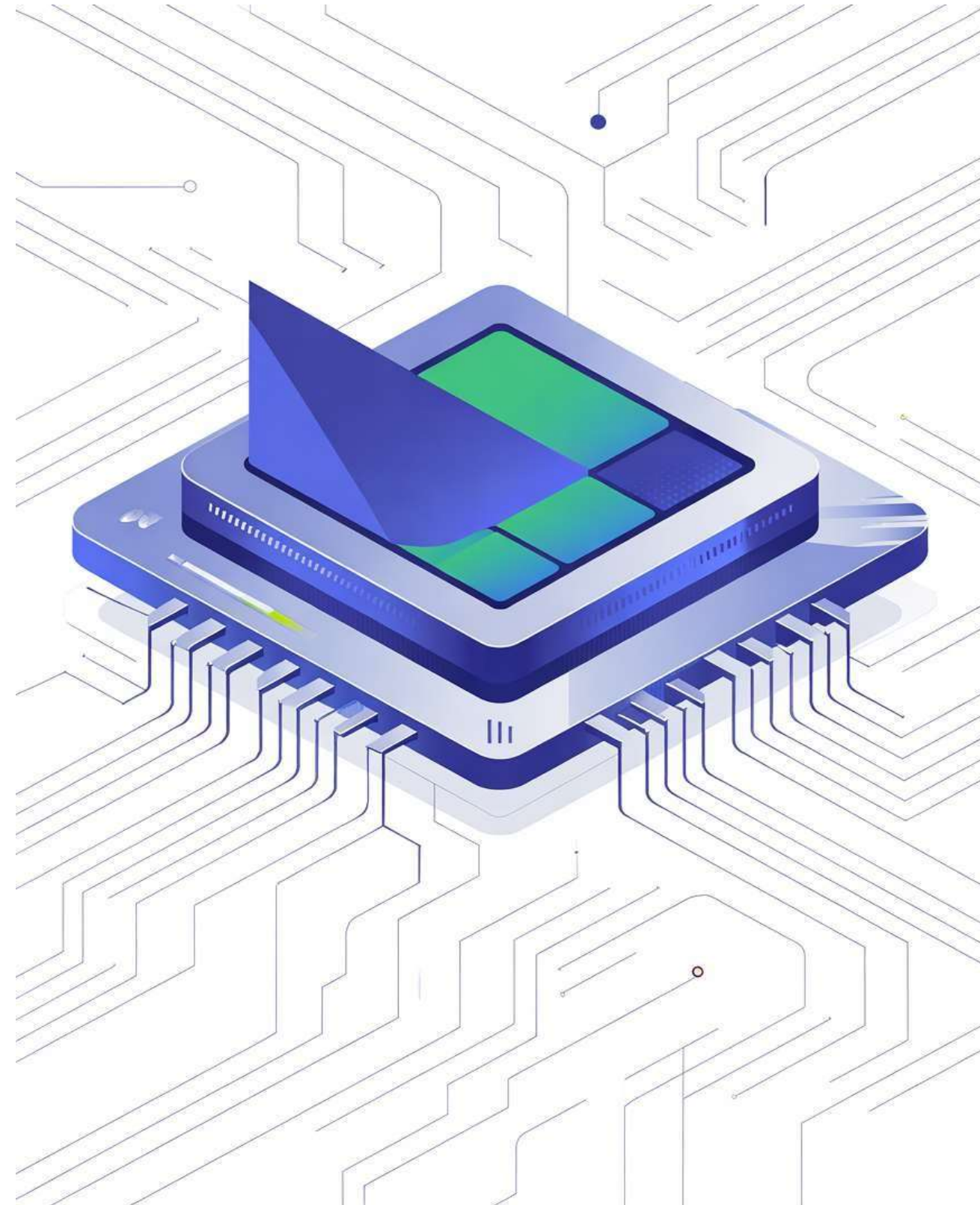


Amplitude Modulation and Demodulation Using Analog Multipliers

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Aim and Objective



To create a modulation and demodulation circuit using analog multipliers



Analyze advantages over the standard envelope detector

Understand advantages over standard diode-based circuits and its utility

Working Principle

THE WORKING OF DEMODULATION :

The circuit receives an AM(Amplitude Modulated) signal and multiplies it with a locally generated carrier signal of the same frequency and phase ,i.e. a coherent signal which is then passed through a low pass filter to obtain back the message signal $[m(t)]$.

01

Input

Receive amplitude modulated signal: $s(t) = [1 + m(t)] \cos(\omega_c t)$ where, $m(t)$ =Message signal and ω_c =Carrier signal frequency

02

Local Carrier

Generate coherent carrier: $V_{LC} = \cos(\omega_o t)$

With the help of the analog multipliers, we obtain the product of the two signals.

The multiplication produces both baseband and high-frequency components:

Then the multiplied signal becomes:

$$s(t) \quad V_{LC} = [1 + m(t)] \cos(\omega_c t) \cos(\omega_c t)$$

Using trigonometric identity: $\cos A \cdot \cos B = 1/2 [\cos(A-B) + \cos(A+B)]$

$$s(t) \quad V_{LC} = \frac{1}{2} [1 + m(t)] + \frac{1}{2} [1 + m(t)] \cos(2\omega_c t)$$

Low Frequency Term

$\frac{1}{2} [1 + m(t)]$ - Baseband message signal

High Frequency Term

$\frac{1}{2} [1 + m(t)] \cos(2\omega_c t)$

On passing this product signal through a low pass filter we can recover the

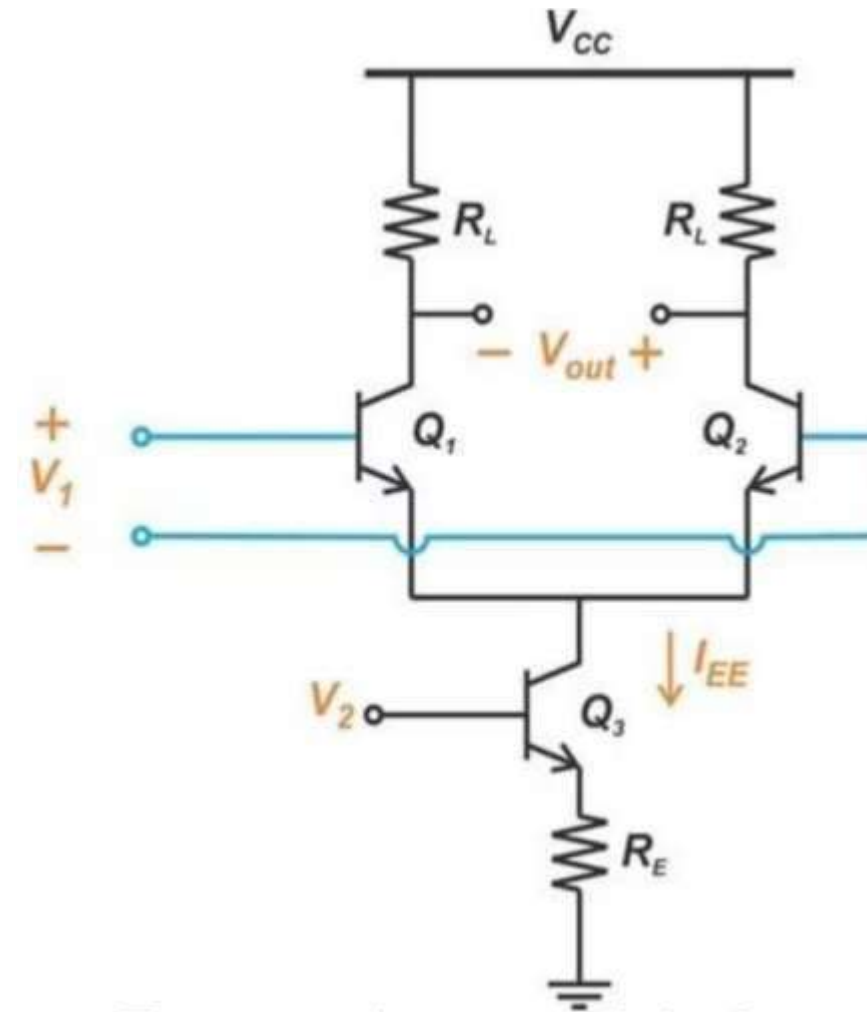
Message signal: $\frac{1}{2} [1 + m(t)]$

Implementation Using Analog Multiplier

For getting the product signal we use the help of analog multipliers.

There are many ways to construct such multipliers, we can use specific IC's (AD633) or we can build discrete circuits like the Gilbert cell.

Now we'll take a look at the implementation of a simplified version of a gilbert cell which is essentially a variable transconductance multiplier.



Through the differential amplifier-based circuit we assume that V_1 is small and I_{EE} is fixed so we can treat it as a constant-gain amplifier.

Since it is in common emitter mode,

$$V_{out} \approx g_m R_L V_1$$

where, $g_m = I_C / V_T \approx I_{EE} / 2V_T$

(Since the circuit is symmetrical, I_{EE} current splits equally from the emitter currents)

Now across R_E , $V_2 \approx I_{EE} \cdot R_E$, (Ignoring drop of 0.7 for high value of (assumption))
substituting for I_{EE} in the expression for g_m we get,

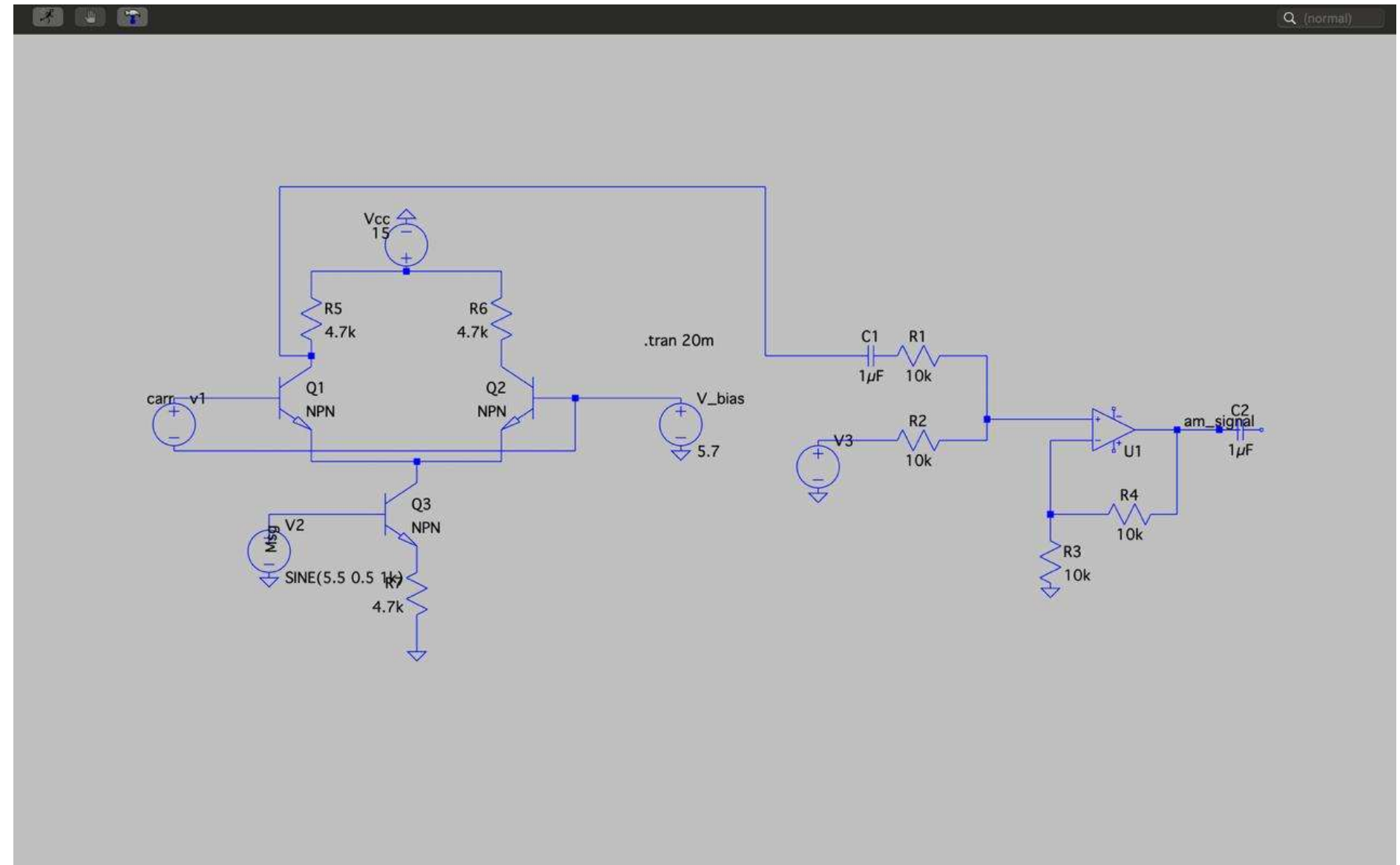
$g_m = V_2 / 2R_E V_T$, substituting in the expression for V_{out} we get,

$V_{out} = (R_L / 2R_E V_T) \times V_1 V_2 \Rightarrow V_{out} = K \cdot V_1 V_2$, which is a product of the two signals.

This is a basic form of a discrete analog multiplier. The Gilbert cell is a modified version of the above multiplier.

Amplitude Modulation Circuit

- The modulation circuit generates AM waves by multiplying signals and adding a local carrier using a non-inverting op-amp adder circuit.



AM Demodulation

The demodulation circuit is of relatively simple task where,

Input

Receive amplitude modulated(AM) signal

Multiplier

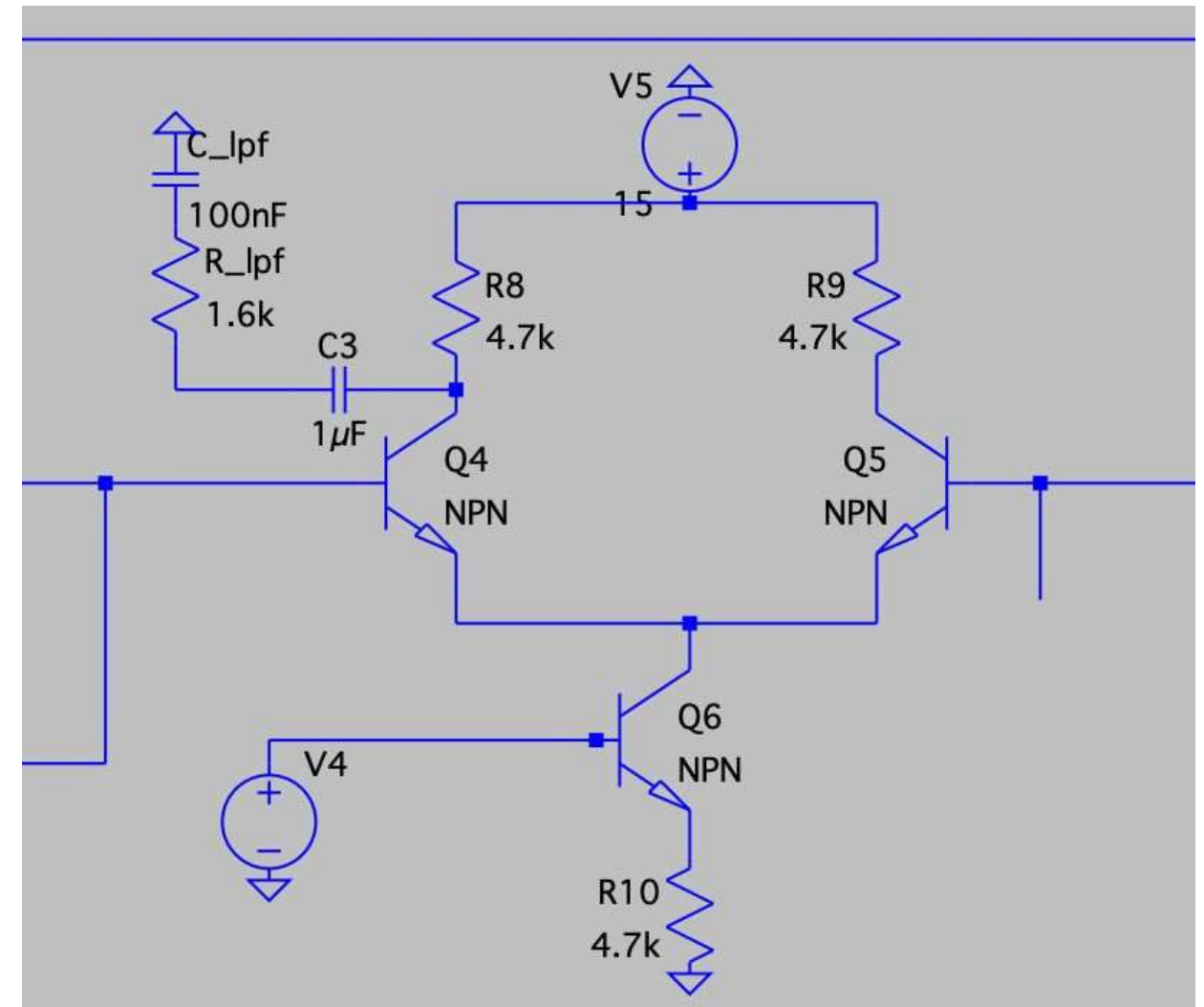
Product with local carrier signal using a multiplier

Low Pass Filter

Passed through a simple low pass filter which removes the high frequency component

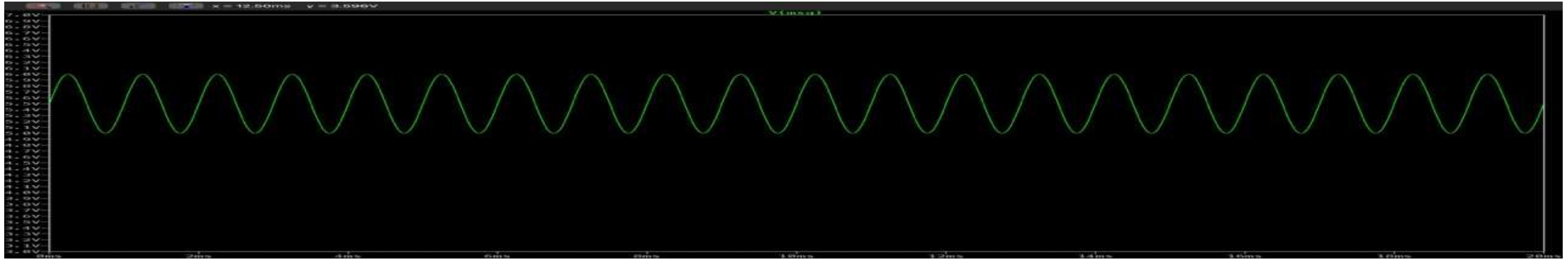
Message Signal

Generates the messenger signal



Demodulation Circuit

RESULT

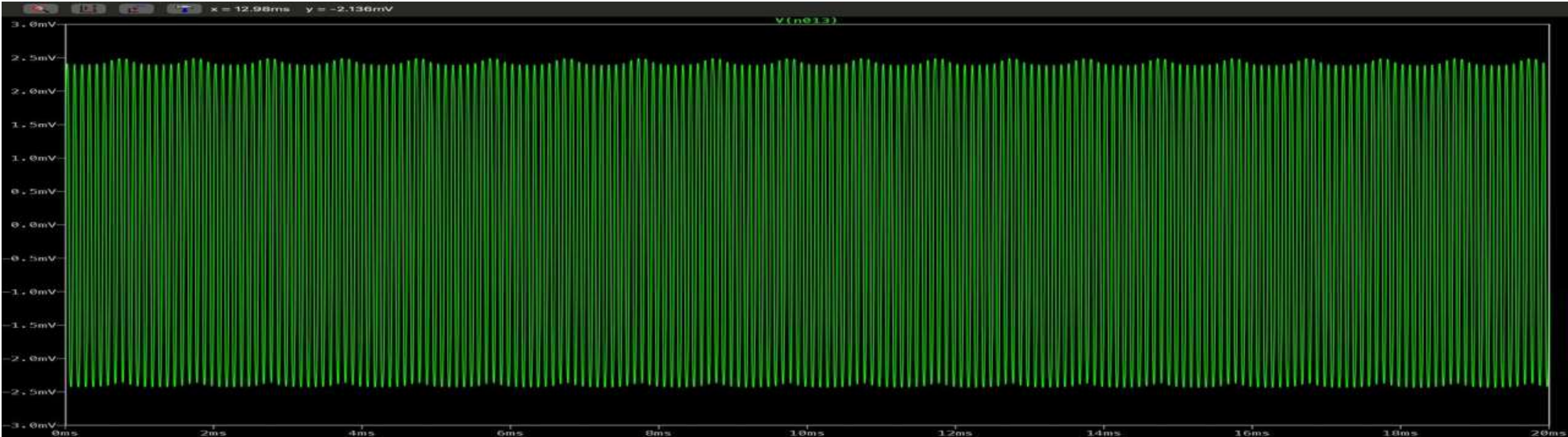


Message Signal

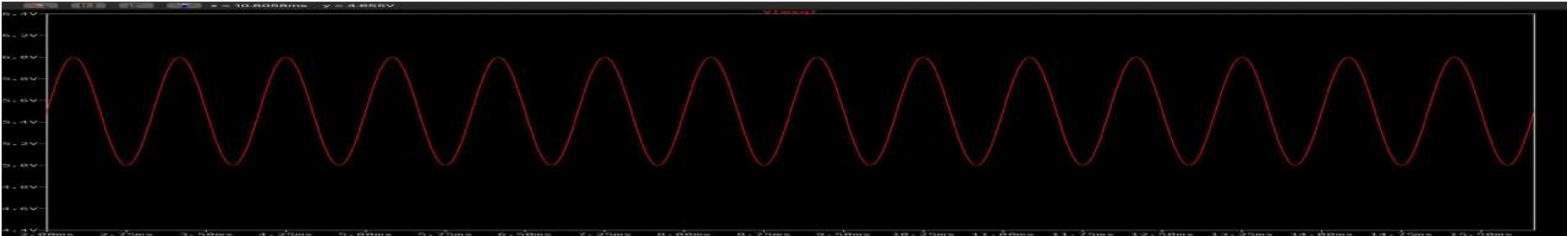


Carrier Signal

AM(Amplitude Modulated Signal



Demodulated Signal



Advantages Over Envelope Detector

Lower Distortion

The Envelope Detector(Diode network) works by nonlinear rectification/biasing, when two signals are applied unwanted terms are produced (DC offsets,harmonics,distortion etc.)

The distortion is minimized a lot when using a multiplier since it linearizes the transistor behaviour(using differential amps)

Signal Versatility

Product detector works with AM,DSB-SC(Double sideband suppressed carrier), and SSB signals.

Envelope detector only handles AM signals.

Superior Performance

More reliable signal recovery with better fidelity across various modulation schemes. Supports mixing(frequency translation)

Frequency range of operation is in the order of GHz for Gilbert cells which is much lesser for the basic modulators.

Real-World Applications



RF Systems & Mixers

Used in telecommunications for frequency mixing and phase-locked loops (PLLs) for stable frequency generation in radios and cellular systems.



True RMS Measurement

Enables accurate AC voltage measurement for any waveform shape, not just sine waves, in precision multimeters.



Power Measurement

Analog wattmeters multiply instantaneous voltage and current signals for accurate power readings.

Ways To Improve



[AD534, Four-Quadrant Precision Multiplier, Analog Devices, circa 1976 - CHM Revolution](#)

Four-Quadrant Translinear Multiplier

The AD534 IC employs a modified Gilbert cell that multiplies currents instead of voltages, offering wideband operation with fewer limitations than basic transconductance multipliers.

Enhanced Bandwidth

Operates across wider frequency ranges

Better Linearity

Improved accuracy in signal multiplication

Lower Noise

Reduced interference and distortion